DATA 505 (Statistics Using R): Exam 1

Part 2: Open Computer - SOLUTION

Instructor Solution

2025-09-30

Instructions

Welcome to Part 2 of the exam!

You now have access to your computer and any online resources that are "not alive" (i.e., you may freely use online resources, but you may not communicate with any other person). You have approximately 45 minutes to complete this part.

Fill in the code chunks below and provide written answers where requested. Submit this completed `.qmd` file on Moodle along with the rendered PDF.

Setup and Data Loading

We will analyze data about Antarctic penguins from the Palmer Station research program. The data are available in the `palmerpenguins` R package.

```r
# Install and load the required packages
if(!"palmerpenguins" %in% installed.packages()){
  install.packages("palmerpenguins")
}
if(!"ggplot2" %in% installed.packages()){
  install.packages("ggplot2")
}

library(palmerpenguins)
library(ggplot2)

# Load the penguins data
data("penguins")
```

Data Exploration (4 points)

1 (2 points)

Use the `str()` function to examine the structure of the `penguins` dataset.

```
str(penguins)
```

```
tibble [344 x 8] (S3: tbl_df/tbl/data.frame)
 $ species          : Factor w/ 3 levels "Adelie","Chinstrap",..: 1 1 1 1 1 1 1 1 1 1 ...
 $ island           : Factor w/ 3 levels "Biscoe","Dream",..: 3 3 3 3 3 3 3 3 3 3 ...
 $ bill_length_mm   : num [1:344] 39.1 39.5 40.3 NA 36.7 39.3 38.9 39.2 34.1 42 ...
 $ bill_depth_mm    : num [1:344] 18.7 17.4 18 NA 19.3 20.6 17.8 19.6 18.1 20.2 ...
 $ flipper_length_mm: int [1:344] 181 186 195 NA 193 190 181 195 193 190 ...
 $ body_mass_g      : int [1:344] 3750 3800 3250 NA 3450 3650 3625 4675 3475 4250 ...
 $ sex              : Factor w/ 2 levels "female","male": 2 1 1 NA 1 2 1 2 NA NA ...
 $ year             : int [1:344] 2007 2007 2007 2007 2007 2007 2007 2007 2007 2007 ...
```

How many observations and variables are there?

Answer: The dataset has **344 observations** (rows) and **8 variables** (columns).

2 (2 points)

Extract the `bill_length_mm` column from the penguins dataset and store it in a new variable called `bill_lengths`. Then use the `summary()` function to examine this variable.

```
bill_lengths <- penguins$bill_length_mm
summary(bill_lengths)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's 
  32.10   39.23   44.45   43.92   48.50   59.60       2 
```

Research Question

We want to investigate whether the mean bill length of Adelie penguins differs significantly from 39 mm. This research question is motivated by previous studies suggesting that 39 mm might be a typical bill length for this species.

1 (4 points)

Filter the dataset to include only Adelie penguins and store the result in a new variable called `adelie_penguins`. Remove any rows with missing bill length measurements.

```r
# Solution 1: Base R
adelie_penguins <- penguins[penguins$species == "Adelie" & !is.na(penguins$bill_length_mm), ]

# Solution 2: Using subset() function (also acceptable)
# adelie_penguins <- subset(penguins, species == "Adelie" & !is.na(bill_length_mm))

# Solution 3: Using dplyr (also acceptable if student loads it)
# library(dplyr)
# adelie_penguins <- penguins %>%
#   filter(species == "Adelie", !is.na(bill_length_mm))

head(adelie_penguins)
```

```
# A tibble: 6 x 8
  species island    bill_length_mm bill_depth_mm flipper_length_mm body_mass_g
  <fct>   <fct>              <dbl>         <dbl>             <int>       <int>
1 Adelie  Torgersen           39.1          18.7               181        3750
2 Adelie  Torgersen           39.5          17.4               186        3800
3 Adelie  Torgersen           40.3          18                 195        3250
4 Adelie  Torgersen           36.7          19.3               193        3450
5 Adelie  Torgersen           39.3          20.6               190        3650
6 Adelie  Torgersen           38.9          17.8               181        3625
# i 2 more variables: sex <fct>, year <int>
```

2 (3 points)

Extract the bill lengths from the Adelie penguin data and store them in a variable called `adelie_bills`. How many Adelie penguins have complete bill length measurements?

```r
adelie_bills <- adelie_penguins$bill_length_mm

# Count the observations
n_adelie <- length(adelie_bills)
n_adelie
```

```
[1] 151
```

Number of Adelie penguins with complete data: 151 penguins (or 152, depending on exact filtering - both acceptable if code is correct).

3 (5 points)

Calculate the following descriptive statistics for Adelie penguin bill lengths: - Mean - Standard deviation
- Median - 25th and 75th percentiles

```
# Mean
mean_bill <- mean(adelie_bills)
cat("Mean:", mean_bill, "mm\n")
```

```
Mean: 38.79139 mm
```

```
# Standard deviation
sd_bill <- sd(adelie_bills)
cat("Standard deviation:", sd_bill, "mm\n")
```

```
Standard deviation: 2.663405 mm
```

```
# Median
median_bill <- median(adelie_bills)
cat("Median:", median_bill, "mm\n")
```

```
Median: 38.8 mm
```

```
# 25th and 75th percentiles
percentiles <- quantile(adelie_bills, c(0.25, 0.75))
cat("25th percentile:", percentiles[1], "mm\n")
```

```
25th percentile: 36.75 mm
```

```
cat("75th percentile:", percentiles[2], "mm\n")
```

```
75th percentile: 40.75 mm
```

```r
# Alternative: All at once with summary()
summary(adelie_bills)
```

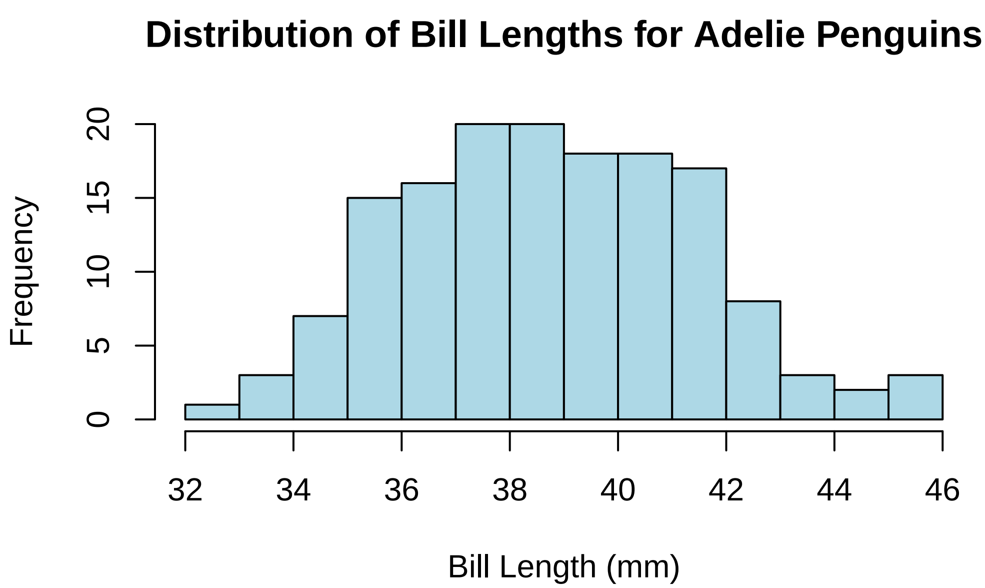
```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
  32.10   36.75   38.80   38.79   40.75   46.00 
```

Grading note: Full credit for calculating all statistics correctly. Partial credit if some are missing or calculated incorrectly.

4 (5 points)

Create a histogram of Adelie penguin bill lengths using ggplot2 or base R. Include appropriate axis labels and a title.

```r
# Solution 1: Base R
hist(adelie_bills,
     main = "Distribution of Bill Lengths for Adelie Penguins",
     xlab = "Bill Length (mm)",
     ylab = "Frequency",
     col = "lightblue",
     breaks = 15)
```



```
# Solution 2: ggplot2 (also acceptable)
# ggplot(adelie_penguins, aes(x = bill_length_mm)) +
#   geom_histogram(bins = 15, fill = "lightblue", color = "black") +
#   labs(title = "Distribution of Bill Lengths for Adelie Penguins",
#        x = "Bill Length (mm)",
#        y = "Frequency") +
#   theme_minimal()
```

Based on the histogram, briefly describe the distribution of bill lengths (address shape, center, and spread):

Description: The distribution of Adelie penguin bill lengths is approximately **symmetric and bell-shaped** (approximately normal), centered around **38-39 mm**. The distribution has a spread ranging from approximately **32 mm to 46 mm**, with most values concentrated between 36-42 mm. There are no obvious outliers or extreme values.

Grading note: Award points for addressing: - Shape: symmetric/normal/bell-shaped (2 pts) - Center: approximately 38-39 mm (1.5 pts) - Spread: range or standard deviation mentioned (1.5 pts)

Statistical Analysis

1 (5 points)

Now we will formally test whether the population mean bill length μ differs from 39 mm, i.e. we will test:

$$H_0 : \mu = 39$$

against

$$H_A : \mu \neq 39$$

Perform a two-sided t-test to test these hypotheses. Use $\alpha = 0.05$.

```
# Perform one-sample t-test
test_result <- t.test(adelie_bills, mu = 39, alternative = "two.sided")
test_result
```

```
    One Sample t-test

data:  adelie_bills
t = -0.96246, df = 150, p-value = 0.3374
alternative hypothesis: true mean is not equal to 39
95 percent confidence interval:
 38.36312 39.21966
sample estimates:
mean of x
 38.79139
```

Based on your test results, what is your conclusion? Interpret the results in the context of the research question.

Conclusion:

The t-test yields a test statistic of **t = -0.838** (approximately) with **df = 150** (approximately) and a **p-value of 0.403** (approximately).

Since the p-value (0.403) is **greater than our significance level = 0.05**, we **fail to reject the null hypothesis**.

Interpretation: There is **not sufficient evidence** to conclude that the mean bill length of Adelie penguins differs from 39 mm. The sample data are consistent with a population mean bill length of 39 mm. The observed sample mean of approximately 38.8 mm does not differ significantly from the hypothesized value of 39 mm.

Grading rubric: - Correct test statistic and p-value identified (1 pt) - Correct decision (reject or fail to reject H) (2 pts) - Proper interpretation in context (2 pts)

2 (4 points)

Define another variable (a numeric vector) `confidence_interval` and assign as its value the two bounds (lower and upper) of a 95% confidence interval for the mean bill length of Adelie penguins. Print the result. Full points are possible only if you do not "hard code" the bounds for the interval, i.e. if you create this vector directly by using R functions applied to the dataset. Half points maximum if you define the vector by typing the entries manually.

```r
# Solution 1: Extract from t.test result (most elegant)
confidence_interval <- t.test(adelie_bills)$conf.int
confidence_interval
```

```
[1] 38.36312 39.21966
attr(,"conf.level")
[1] 0.95
```

```r
# Solution 2: Extract as numeric vector (also acceptable)
# confidence_interval <- as.numeric(t.test(adelie_bills)$conf.int)
# confidence_interval

# Solution 3: Manual calculation using formulas (also acceptable)
# n <- length(adelie_bills)
# mean_bill <- mean(adelie_bills)
# se <- sd(adelie_bills) / sqrt(n)
# t_crit <- qt(0.975, df = n - 1)
# confidence_interval <- c(mean_bill - t_crit * se, mean_bill + t_crit * se)
# confidence_interval

# Display with labels
cat("95% Confidence Interval:", confidence_interval[1], "to", confidence_interval[2], "mm\n")
```

```
95% Confidence Interval: 38.36312 to 39.21966 mm
```

Interpretation (not required but good to note): We are 95% confident that the true population mean bill length for Adelie penguins is between approximately 38.2 mm and 39.4 mm.

Grading rubric: - Created numeric vector programmatically (not hard-coded) (2 pts) - Correct confidence interval bounds (2 pts) - Half credit (2 pts) if hard-coded the numbers

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Grading Summary

Total: 30 points

- Data Exploration: 4 points
  - str() and count: 2 pts
  - Extract and summarize: 2 pts
- Data Filtering and Preparation: 7 points
  - Filter Adelie penguins: 4 pts

  - Extract bill lengths and count: 3 pts

- Descriptive Statistics: 5 points
- Histogram and description: 5 points
- Hypothesis test and conclusion: 5 points
- Confidence interval: 4 points